Polycythemia (ERYTHROCYTOSIS)

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LEARNING OBJECTIVES

- At the end of the lecture ,Students should be able to
- Define Polycythemia
- Discuss its different types
- Explain the pathophysiology of the types of polycythemia
- Enlist the different Risk factors .

Definition

- Polycythemia (erythrocytosis) refers to an abnormal increase in the red blood cell (RBC) count, hemoglobin concentration, or hematocrit in the blood above the upper limit of normal for the patient's age and sex.
- Polycythemia is a condition where hematocrit and RBC mass increase above normal levels, leading to increased blood viscosity.

Normal RBC Function

- Major function of RBCs is the transport of respiratory gases.
- Following are the functions of RBCs:
- 1. **Transport of Oxygen from the Lungs to the Tissues**
  Hemoglobin in RBC combines with oxygen to form oxyhemoglobin. About 97% of oxygen is transported in blood in the form of oxyhemoglobin .
- 2. **Transport of Carbon Dioxide from the Tissues to the Lungs**
  Hemoglobin combines with carbon dioxide and form carbhemoglobin. About 30% of carbon dioxide is transported in this form.
- Produced in bone marrow.
- Controlled mainly by erythropoietin (EPO)
- Kidneys regulate RBC production depending on oxygen levels

Role of Erythropoietin

- Most important general factor for erythropoiesis is the hormone called erythropoietin. It is also called hemopoietin or erythrocyte stimulating factor.
- Erythropoietin is a glycoprotein with 165 amino acids.
- Major quantity of erythropoietin is secreted by peritubular capillaries of kidney. A small quantity is also secreted from liver and brain.
- **Stimulant for secretion** Hypoxia is the stimulant for the secretion of erythropoietin.
- Erythropoietin causes formation and release of new RBCs into circulation. After secretion, it takes 4 to 5 days to show the action.
- • Increases RBC production in bone marrow
- • Maintains oxygen carrying capacity of blood

Types of Polycythemia

- Depending upon the cause, Polycythemia may be
- 1. Primary Polycythemia (Polycythemia Vera)
- 2. Secondary Polycythemia
- 3. Relative Polycythemia

Patho physiology

- Primary Polycythemia is caused by an abnormality of the cells in the bone marrow that form red blood cells.
- Secondary Polycythemia is caused by a disorder originating outside the bone marrow that causes overstimulation of the normal bone marrow, leading to an overproduction of red blood cells.

Primary Polycythemia (polycythemia Vera)

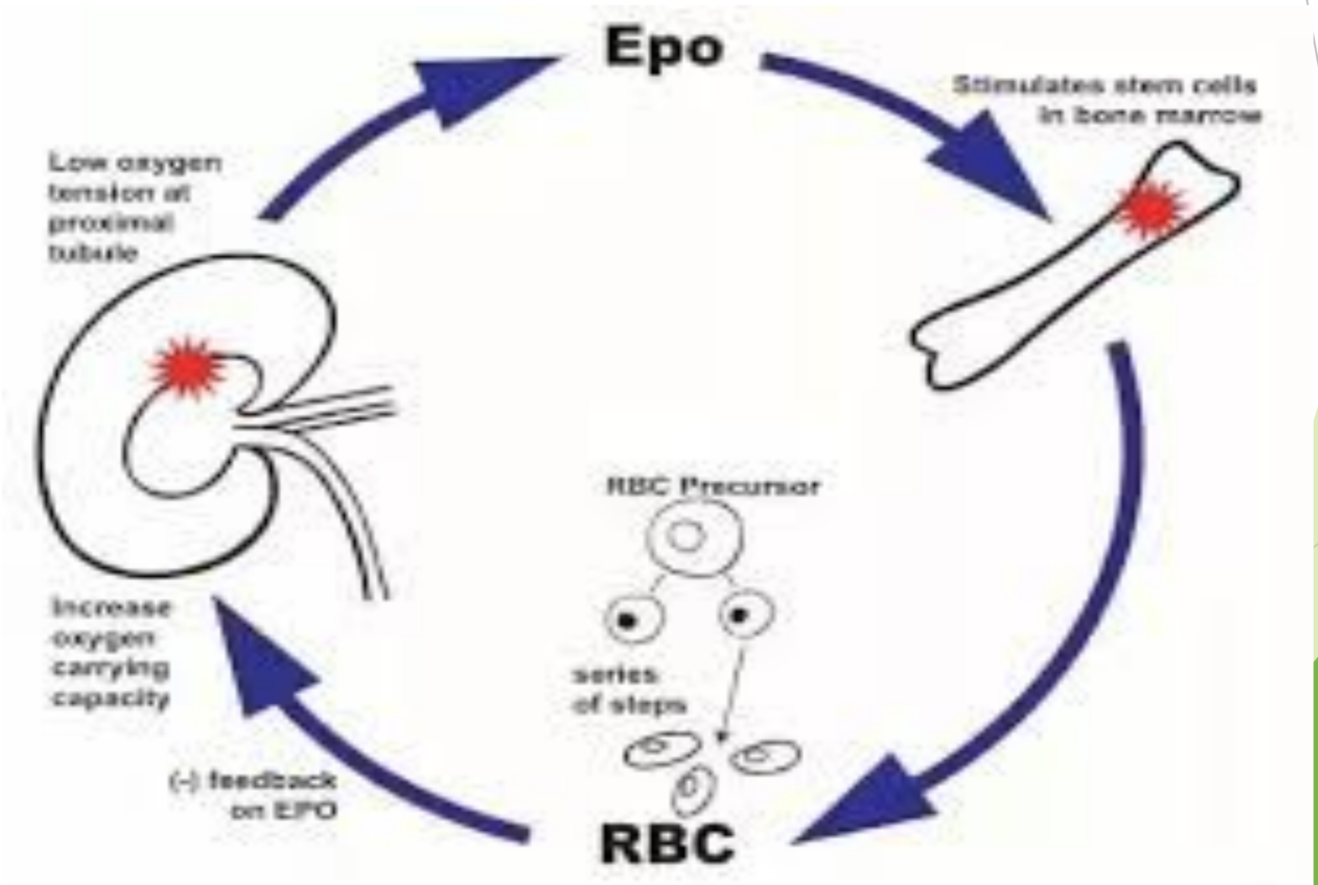
- Primary polycythemia is otherwise known as polycythemia vera. It is a disease characterized by persistent increase in RBC count above 14 million/cu mm of blood. This is always associated with increased white blood cell count above 24,000/cu mm of blood.
- Polycythemia vera occurs in myeloproliferative disorders like malignancy of red bone marrow.
- PV is a type of blood cancer known as myeloproliferative neoplasm, which causes the bone marrow to produce too many RBCs,often with excess WBCs and platelets.
- Most cases of PV are associated with a mutation in the JAK2 gene mutation, leads to uncontrolled cell growth in the bone marrow.
- More common in older age over 50.
- Not dependent on erythropoietin.

Secondary Polycythemia

- Occurs due to increased RBCs production in response to other factors.
- Caused by chronic hypoxia- (COPD, sleep apnea, living at high altitude, congenital heart diseases.)
- Inhabitants of mountains (above 10,000 feet from mean sea level) have an increased RBC count of more than 7 million/cu mm. It is due to hypoxia (decreased oxygen supply to tissues) in high altitude. Hypoxia stimulates kidney to secrete a hormone called erythropoietin. The erythropoietin in turn stimulates the bone marrow to produce more RBCs.
- Erythropoietin(EPO) producing tumors –some kidney tumors can produce erythropoietin.
- Use of anabolic steroids.
- All these conditions lead to hypoxia which stimulates the release of erythropoietin. Erythropoietin stimulates the bone marrow resulting in increased RBC count.

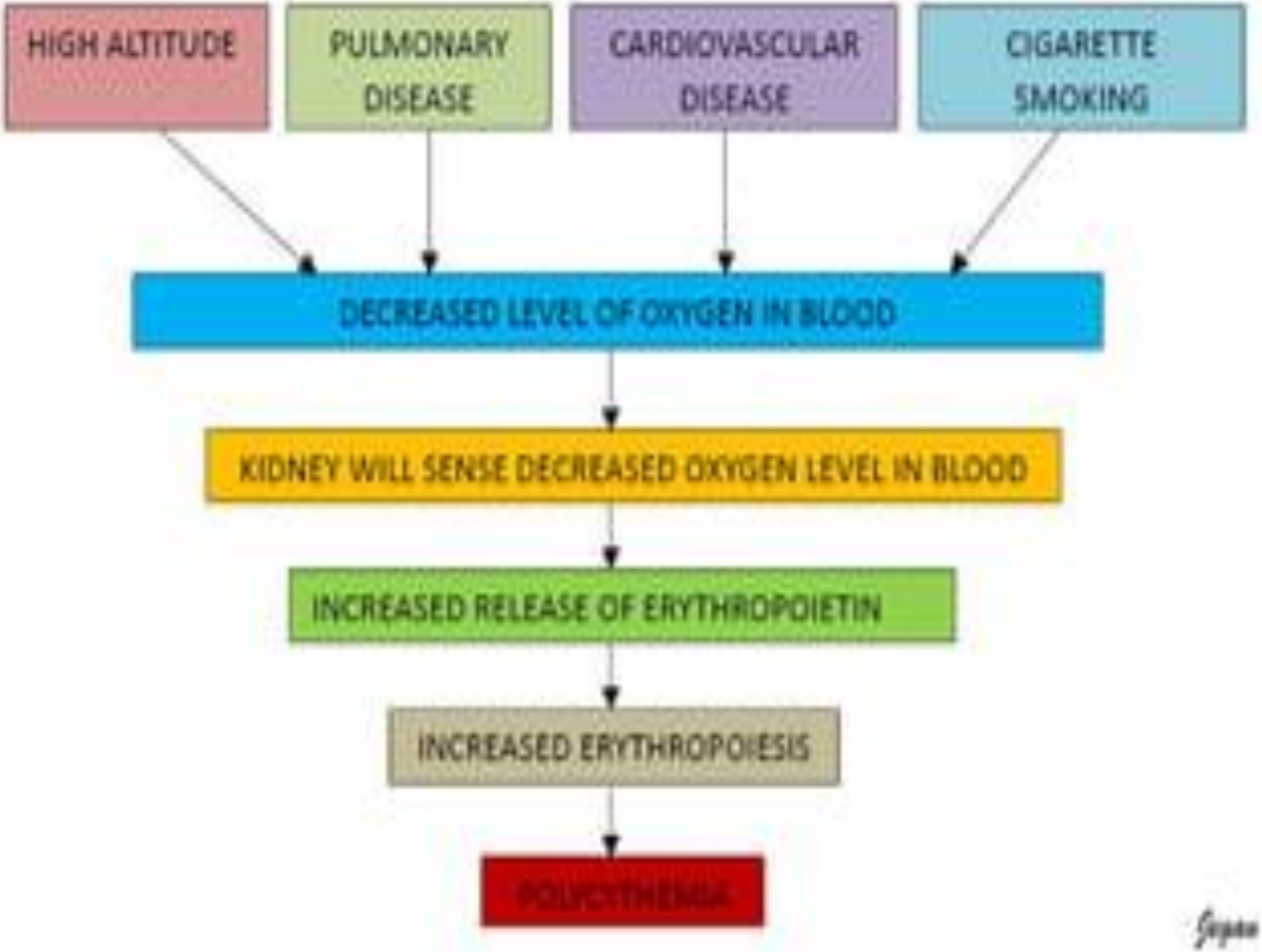
Secondary polycythemia

- Its due to over production of erythropoietin due to chronic hypoxia or EP secreting tumor



Relative Polycythemia

- • RBC number is normal
- Increased concentration of blood due to loss of plasma volume
- • In dehydration (vomiting diarrhea, excessive sweating, use of diuretics)
- • Stress (physical or psychological) obesity,hypertension,smoking,long exposure to carbon monoxide are also risk factors.


HIGH ALTITUDE
PULMONARY DISEASE
CARDIOVASCULAR DISEASE
CIGARETTE SMOKING
DECREASED LEVEL OF OXYGEN IN BLOOD
KIDNEY WILL SENSE DECREASED OXYGEN LEVEL IN BLOOD
INCREASED RELEASE OF ERYTHROPOIETIN
INCREASED ERYTHROPOIESIS
POLYCYTHEMIA

Pathophysiology of Polycythemia Vera

- J AK2 V617F mutation causes a change in signaling pathway that normally regulates a blood cell production.

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- Bone marrow hyperplasia and increased blood cell precursors (myeloid progenitor stem cells) in the bone marrow.

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- Uncontrolled production of RBCs and sometimes WBCs and blood platelets

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- Increased red cells mass makes the blood thicker, which can slow blood flow and increase the risk of clotting.

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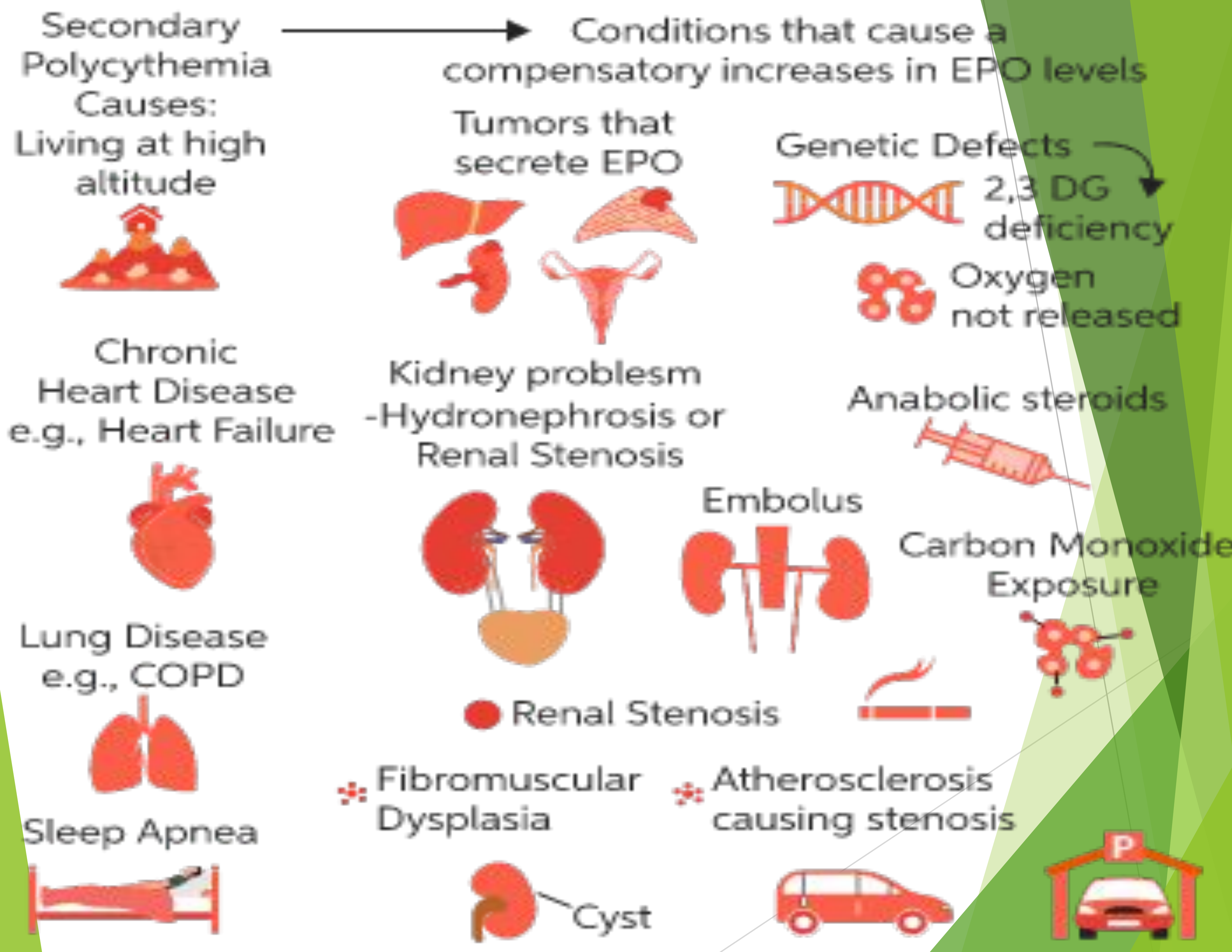
- Splenomegaly occurs because of increased workload on the spleen to filter and manage excess blood cells.


Secondary Polycythemia Causes:
Conditions that cause a compensatory increases in EPO levels
Living at high altitude
Tumors that secrete EPO
Genetic Defects
2,3 DG deficiency
Oxygen not released
Chronic Heart Disease e.g., Heart Failure
Kidney problesm -Hydronephrosis or Renal Stenosis
Anabolic steroids
Embolus
Carbon Monoxide Exposure
Lung Disease e.g., COPD
Renal Stenosis
Sleep Apnea
Fibromuscular Dysplasia
Atherosclerosis causing stenosis
Cyst

Clinical features (Hyperviscosity Syndrome)

Symptoms

- Headache
- Fatigue
- Pruritus
- Dizziness
- Visual disturbances
- Weight loss
- Erythromelalgia
- Dyspnea
- Joint symptoms
- Epigastric discomfort
- thrombosis

Signs

- Splenomegaly
- Skin plethora (Increased red cell volume)
- Conjunctival plethora
- Engorged vessels in the optic fluid
- Hepatomegaly

Laboratory test

- CBC
- Peripheral blood smear
- Bone marrow examination
- EPO level

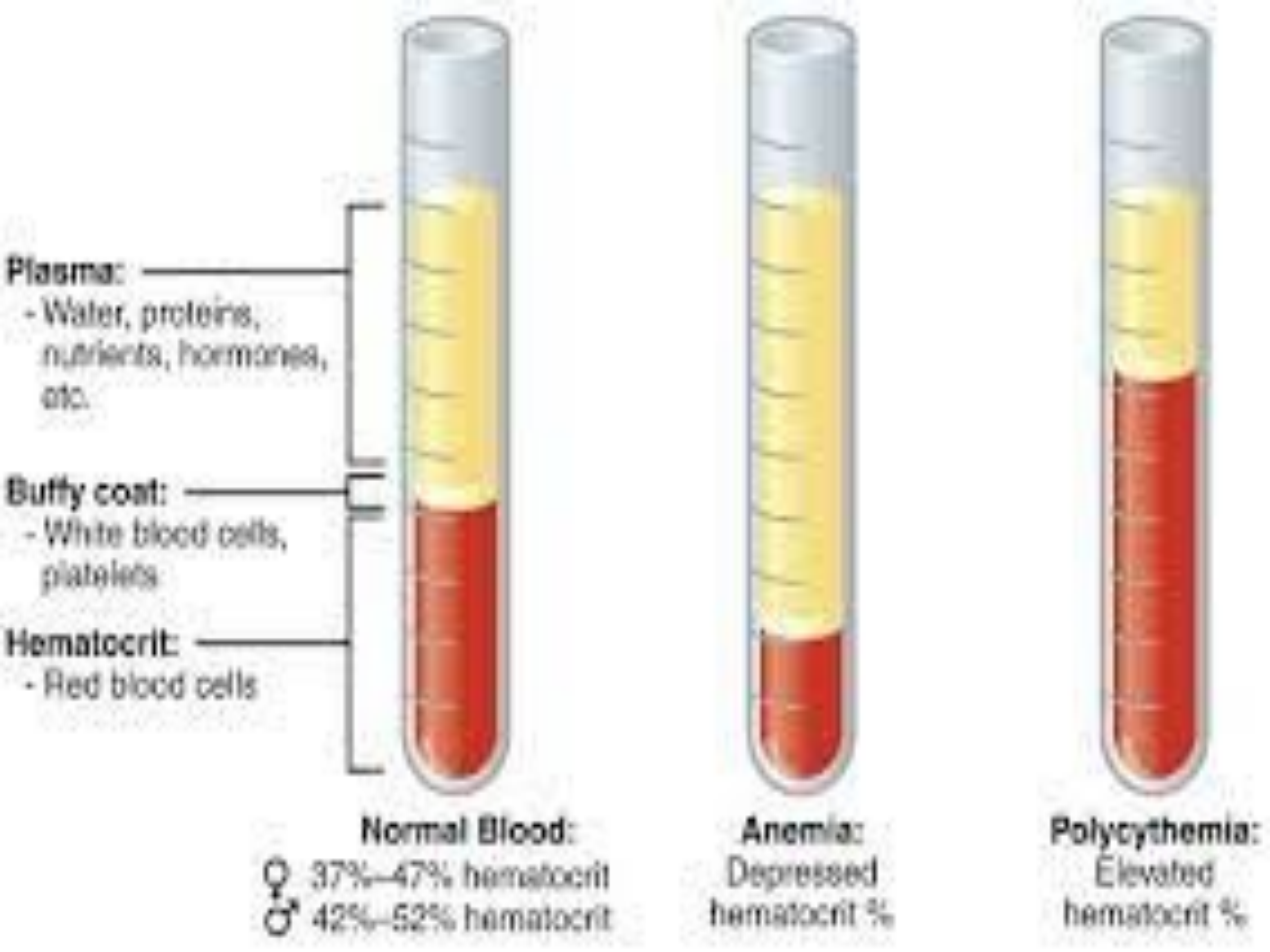
- **Red cell**
- Increase Hct, Hb

Not increase in PV with iron deficiency

- **White cell**

Leukocytosis

- Increase basophil, eosinophil
- **Platelet**
- Increase or normal


Plasma:
- Water, proteins, nutrients, hormones, etc.
Buffy coat:
- White blood cells, platelets
Hematocrit:
- Red blood cells
Normal Blood:
♀ 37%–47% hematocrit
♂ 42%–52% hematocrit
Anemia:
Depressed hematocrit %
Polycythemia:
Elevated hematocrit %

Blood smear

RBC excess red cells,

- hypochromic microcytic red cells (iron deficiency)
- WBC increase with band form, myelocyte ,metamyelocyte
- Platelet increase

Bone marrow

- Hypercellular marrow
- Increase erythroid, myeloid, megakaryocyte (panmyelosis)
- Normal maturation of myeloid series
- ME 2-31
- Increase megakaryocytes with different size

Cause of death

- Thrombosis
- Malignancy acute leukemia, non-RE malignancy
- Myelofibrosis
- Bleeding

MANAGEMENT

- LIFE STYLE MODIFICATON
- STOP SMOKING
- EXERSICE
- MEDICATION

"Self-belief and hard work will always earn you success."

Virat Kohli